

Comeback

Input file: **standard input**
Output file: **standard output**
Time limit: **1 second**
Memory limit: **256 megabytes**

Clarence is a producer on the hit TV show *Survival*. In the newest season, Amanda and Bob are the final two contestants remaining. However, due to some logistical challenges, it has not been possible to reconvene the other contestants to vote for a winner. It has therefore been ruled that the title of Sole Survivalist and the billion dollar prize will be decided by an arm-wrestling match.

The arm-wrestling match consists of n rounds. In each round, either Amanda wins (denoted 'A') or Bob wins (denoted 'B'). At the end of the match, whichever contestant wins the most rounds will be declared the winner overall.

Clarence has the unenviable job of making good TV from this dismal content. However, he realises that in the original footage, the host did not specify how many rounds there would be in the match! Therefore, Clarence has the freedom to edit out some rounds to make a more compelling show, so long as he does not change the overall winner.

Clarence decides that the narrative shall be a comeback victory. First, the eventual loser will win some k rounds in a row, but then the eventual winner will win $k + 1$ rounds to clinch the victory. Any other rounds will be cut out of the final edit.

However, Clarence would also like the episode to contain a lot of suspenseful music and dramatic reactions, so the number of rounds included in the broadcast should be as large as possible.

Help Clarence decide which rounds to include or exclude for the best comeback story.

Input

The first line of input consists of a single integer n ($1 \leq n < 200,000$, n is odd), representing the number of rounds.

The second line of input consists of a string of n characters, the i th of which is either:

- 'A', representing that Amanda won the i th round, or
- 'B', representing that Bob won the i th round.

Output

The first line of output should contain the maximum number of rounds that can be included in the broadcast, while satisfying the comeback condition.

The second line of input should contain a string of n characters, the i th of which is either:

- 'A', representing that Amanda won the i th round, and it is to be included in the broadcast,
- 'B', representing that Bob won the i th round, and it is to be included in the broadcast, or
- '_', representing that the i th round is to be excluded from the broadcast.

If there are several possible ways to choose rounds for inclusion or exclusion, you may output any of them.

Examples

standard input	standard output
5 ABABA	3 _BA_A
11 BAAABBBAAA	5 B___B___AAA
11 BBBAAAAABBB	5 ___AA___BBB
1 A	1 A

Note

In sample case 1, the broadcast has one win for Bob followed by two wins for Amanda. This is the biggest comeback that can be achieved.

In sample case 2, the broadcast has two wins for Bob followed by three wins for Amanda. This is the biggest comeback that can be achieved.

In sample case 3, the string 'BBBAAAA___' would be a larger comeback, but the original taping had more wins for Bob (6) than Amanda (5), so '___AA___BBB' is the largest comeback that can be achieved *while maintaining the eventual winner*.

In sample case 4, the match consists of just one round, which Clarence will have to air as is. This is considered a valid one-nil comeback.